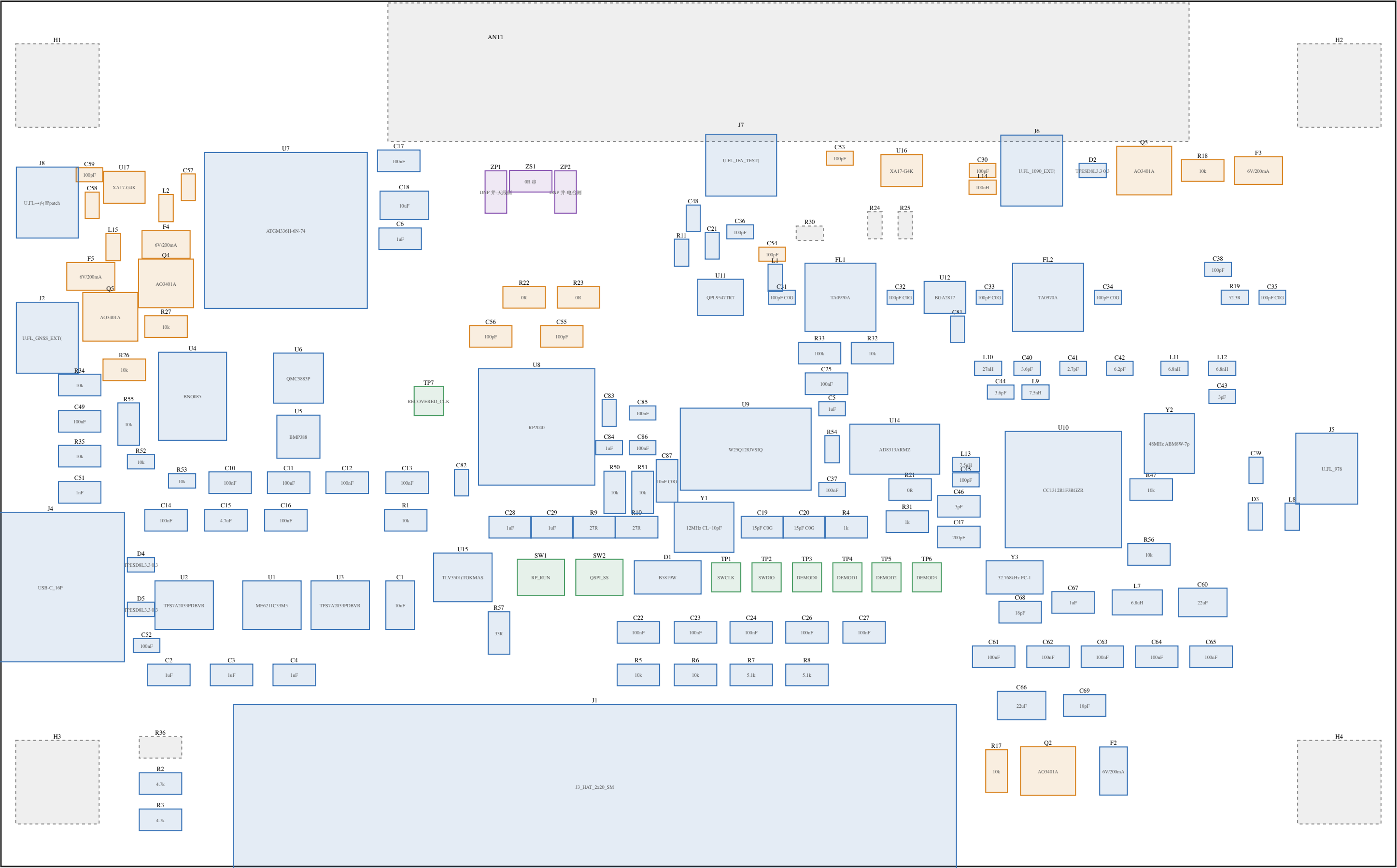


Expansion Board V3.9 Assembly Map

Single-sided: all parts on F.Cu, back side empty; dashed = not populated



Designator index

ANT1	IFA_1090	(85.5,57.8) 0°	C81	470pF C0G	(118.6,73.5) 90°	R31	1k	(115.0,87.3) 0°
C1	10uF	(78.6,93.3) -90°	C82	100nF	(83.0,84.5) -90°	R32	10k	(112.5,75.2) 0°
C2	1uF	(62.0,98.3) 180°	C83	100nF	(93.6,79.5) -90°	R33	100k	(108.7,75.2) 180°
C3	1uF	(66.5,98.3) 0°	C84	1uF	(93.6,82.0) 0°	R34	10k	(55.6,77.5) 0°
C4	1uF	(71.0,98.3) 0°	C85	100nF	(96.0,79.5) 0°	R35	10k	(55.6,82.6) 0°
C5	1uF	(109.6,79.2) 0°	C86	100nF	(96.0,82.0) 0°	R36	1k DNP	(61.4,103.5) 180°
C6	1uF	(78.6,67.0) 0°	C87	10nF C0G	(97.8,84.4) 90°	R47	10k	(132.5,85.0) 0°
C10	100nF	(66.4,84.5) 0°	D1	B5819W	(97.8,91.3) 0°	R50	10k	(94.0,85.2) 90°
C11	100nF	(70.6,84.5) 0°	D2	TPESD8L3.3 0.3pF	(128.3,62.1) 0°	R51	10k	(96.0,85.2) -90°
C12	100nF	(74.8,84.5) 0°	D3	TPESD8L3.3 0.3pF	(140.0,86.9) -90°	R52	10k	(60.0,83.0) 0°
C13	100nF	(79.1,84.5) 0°	D4	TPESD8L3.3 0.3pF	(60.0,90.4) 180°	R53	10k	(63.0,84.4) 180°
C14	100nF	(61.8,87.2) 0°	D5	TPESD8L3.3 0.3pF	(60.0,93.6) 180°	R54	10R	(109.6,82.1) 90°
C15	4.7uF	(66.1,87.2) 180°	F2	6V/200mA	(129.8,105.2) -90°	R55	10k	(59.1,80.3) 90°
C16	100nF	(70.4,87.2) 0°	F3	6V/200mA	(140.2,62.1) 0°	R56	10k	(132.3,89.7) 0°
C17	100nF	(78.5,61.4) 0°	F4	6V/200mA	(61.8,67.4) 0°	R57	33R	(85.7,95.3) -90°
C18	10uF	(78.9,64.6) 180°	F5	6V/200mA	(56.4,69.7) 0°	SW1	RP_RUN	(88.7,91.3) 0°
C19	15pF C0G	(104.6,87.7) 0°	FL1	TA0970A	(110.2,71.2) 0°	SW2	QSPL_SS	(92.9,91.3) 0°
C20	15pF C0G	(107.6,87.7) 180°	FL2	TA0970A	(125.1,71.2) 0°	TP1	SWCLK	(102.0,91.3) 0°
C21	100pF	(101.0,67.5) 90°	H1	MountingHole_2.7	(54.0,56.0) 0°	TP2	SWDIO	(104.9,91.3) 0°
C22	100nF	(95.7,95.2) 0°	H2	MountingHole_2.7	(146.0,56.0) 0°	TP3	DEMODO	(107.8,91.3) 0°
C23	100nF	(99.8,95.2) 0°	H3	MountingHole_2.7	(54.0,106.0) 0°	TP4	DEMOD1	(110.7,91.3) 0°
C24	100nF	(103.8,95.2) 0°	H4	MountingHole_2.7	(146.0,106.0) 0°	TP5	DEMOD2	(113.5,91.3) 0°
C25	100nF	(109.2,77.4) 0°	J1	J3_HAT_2x20_SMD排	(92.6,106.3) 90°	TP6	DEMOD3	(116.4,91.3) 0°
C26	100nF	(107.8,95.2) 0°	J2	U.FL_GNSS_EXT(经尾	(52.9,74.1) 180°	TP7	RECOVERED_CLK	(80.7,78.7) 0°
C27	100nF	(111.9,95.2) 0°	J4	USB-C_16P	(53.5,92.0) -90°	U1	ME6211C33M5	(69.4,93.3) 180°
C28	1uF	(86.5,87.7) 0°	J5	U.FL_978	(145.5,83.5) 0°	U2	TPS7A2033PDBVR	(63.1,93.3) 180°
C29	1uF	(89.5,87.7) 0°	J6	U.FL_1090_EXT(经尾	(124.3,62.1) 0°	U3	TPS7A2033PDBVR	(74.3,93.3) 180°
C30	100pF	(120.4,62.1) 180°	J7	U.FL_IFA_TEST(π后	(103.1,61.3) 90°	U4	BNO085	(63.7,78.3) 0°
C31	100pF C0G	(106.0,71.2) 0°	J8	U.FL→内置patch	(52.9,64.4) 180°	U5	BMP388	(71.3,81.2) 0°
C32	100pF C0G	(114.5,71.2) 0°	L1	18nH 0402CS-18NX	(105.5,69.8) 90°	U6	QMC5883P	(71.3,77.0) 0°
C33	100pF C0G	(120.9,71.2) 0°	L2	33nH	(61.8,64.8) 90°	U7	ATGM336H-6N-74	(70.4,66.4) 0°
C34	100pF C0G	(129.4,71.2) 0°	L7	6.8uH	(131.5,93.1) 0°	U8	RP2040	(88.4,80.5) -90°
C35	100pF C0G	(141.2,71.2) 0°	L8	100nH	(142.6,87.0) 90°	U9	W25Q128JVSIQ	(103.4,82.1) 0°
C36	100pF	(103.0,66.5) 0°	L9	7.5nH	(124.2,78.0) 0°	U10	CC1312R1F3RGZR	(126.2,85.0) 0°
C37	100nF	(109.6,85.0) 0°	L10	27nH	(120.8,76.3) 0°	U11	QPL9547TR7	(101.6,71.2) 0°
C38	100pF	(137.3,69.2) 0°	L11	6.8nH	(134.2,76.3) 0°	U12	BGA2817	(117.7,71.2) 180°
C39	100pF	(140.0,83.6) -90°	L12	6.8nH	(137.6,76.3) 0°	U14	AD8313ARMZ	(114.1,82.1) 0°
C40	3.6pF	(123.6,76.3) 0°	L13	7.5nH	(119.2,83.2) 0°	U15	TLV3501(TOKMAS)	(83.1,91.3) 0°
C41	2.7pF	(126.9,76.3) 0°	L14	100nH	(120.4,63.3) 0°	U16	XA17-G4K	(114.6,62.1) 0°
C42	6.2pF	(130.2,76.3) 180°	L15	33nH	(58.0,67.6) 90°	U17	XA17-G4K	(58.8,63.3) 0°
C43	3pF	(137.6,78.3) 180°	Q2	AO3401A	(125.1,105.2) 0°	Y1	12MHz CL=10pF AB	(100.4,87.7) 0°
C44	3.6pF	(121.7,78.0) 0°	Q3	AO3401A	(132.0,62.1) 180°	Y2	48MHz ABM8W-7pF	(133.8,81.7) 90°
C45	100pF	(119.2,84.3) 180°	Q4	AO3401A	(61.8,70.2) 0°	Y3	32.768kHz FC-135	(122.7,91.3) 0°
C46	3pF	(118.7,86.2) 180°	Q5	AO3401A	(57.8,72.6) 0°	ZP1	DNP 并-天线侧	(85.5,63.6) -90°
C47	200pF	(118.7,88.4) 0°	R1	10k	(79.0,87.2) 180°	ZP2	DNP 并-电台侧	(90.5,63.6) -90°
C48	1uF	(99.6,65.5) 90°	R2	4.7k	(61.4,106.1) 0°	ZS1	0R 串	(88.0,62.9) 0°
C49	100nF	(55.6,80.1) 180°	R3	4.7k	(61.4,108.7) 0°			
C51	1nF	(55.6,85.2) 180°	R4	1k	(110.6,87.7) 0°			
C52	100nF	(60.4,96.2) 0°	R5	10k	(95.7,98.3) 180°			
C53	100pF	(110.2,61.2) 0°	R6	10k	(99.8,98.3) 180°			
C54	100pF	(105.3,68.1) 180°	R7	5.1k	(103.8,98.3) 0°			
C55	100pF	(90.2,74.0) 0°	R8	5.1k	(107.8,98.3) 0°			
C56	100pF	(85.1,74.0) 180°	R9	27R	(92.5,87.7) 180°			
C57	100pF	(63.4,63.3) -90°	R10	27R	(95.6,87.7) 180°			
C58	100pF	(56.5,64.6) 90°	R11	3.32k	(98.8,68.0) 90°			
C59	100pF	(56.3,62.4) 0°	R17	10k	(121.4,105.2) 90°			
C60	22uF	(136.2,93.1) 0°	R18	10k	(136.2,62.1) 0°			
C61	100nF	(121.2,97.0) 180°	R19	52.3R	(138.5,71.2) 0°			
C62	100nF	(125.1,97.0) 0°	R21	0R	(115.2,85.0) 180°			
C63	100nF	(129.0,97.0) 0°	R22	0R	(87.5,71.2) 0°			
C64	100nF	(132.9,97.0) 0°	R23	0R	(91.4,71.2) 180°			
C65	100nF	(136.8,97.0) 0°	R24	0R DNP(旁路外接)	(112.7,66.0) -90°			
C66	22uF	(123.2,100.5) 0°	R25	0R DNP(旁路板载)	(114.8,66.0) -90°			
C67	1uF	(126.9,93.1) 180°	R26	10k	(58.8,76.4) 0°			
C68	18pF	(123.1,93.8) 0°	R27	10k	(61.8,73.3) 0°			
C69	18pF	(127.7,100.5) 0°	R30	1k DNP	(108.0,66.6) 180°			